

EXERCISE 02

SOL FOUNDATIONS

SQL Aggregate Functions & Operations

Q1) Answer: SELECT DISTINCT department
FROM student;

output

department
IT
HR
Finance

Q2) Answer: SELECT department, AVG(age) AS avg-age
FROM students
GROUP BY department

output

department	avg-age
IT	20.5
HR	22.0
Finance	23.0

Q3) Answer: SELECT department, COUNT(*) AS student-count
FROM
GROUP BY department
HAVING COUNT(*) > 1;

output

department	student-count
IT	2
HR	2

Q4 Answer: `SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;`

output

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q5 Answer: `SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR')
AND age > 21;`

output

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

The Courses Table

Q6 Answer: `SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;`

output:

department	total credits
IT	11

Q7 Answer: `SELECT course_id, course_name, department, credits
FROM courses
WHERE credits < 4;`

output

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Q8 Answer: `SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;`

output

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

The Enrollments Table

Q9 Answer: `SELECT
MAX(grade) AS max_grade;
MIN(grade) AS min_grade;
AVG(grade) AS avg_grade;
FROM enrollments;`

output

Max_grade	Min_grade	avg_grade
90	78	84.6

⑩ Answer: SELECT Course_id, COUNT(*) AS enrolment_count
FROM enrollments
GROUP BY Course_id;

Output

Course_id	enrolment_count
101	1
102	1
103	1
104	1
105	1

The Salaries Table

⑪ Answer: SELECT
Department,
SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus,
FROM Salaries
GROUP BY Department;

Output

Department	total_salary	total_bonus
IT	122000	10500
HR	109000	7800
Finance	70000	6000

⑫ Answer: SELECT
Department,
AVG(salary) AS avg_salary
FROM Salaries
GROUP BY Department
HAVING AVG(salary) > 8000;

Output:

Department	avg-salary
IT	61000
Finance	70000

13) Answer: SELECT

employee-id,

name,

salary,

bonus,

(salary + bonus) AS total-compensation

FROM salaries

WHERE (salary + bonus) > 60000;

output:

employee-id	name	salary	bonus	total-compensation
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

The Project Table

14) Answer: SELECT

Department,

SUM(budget) AS total-budget,

AVG(budget) AS avg-budget.

FROM projects

GROUP BY Department

HAVING AVG(budget) > 70000

Output:

Department	total-budget	avg-budget
IT	27000	135000
Finance	80000	80000

15) Answer: SELECT

Project-id,
Project-name,
Department,
Budget

FROM projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';

Output:

Project-id	Project-name	Department	Budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
3	HR Portal	HR	50000